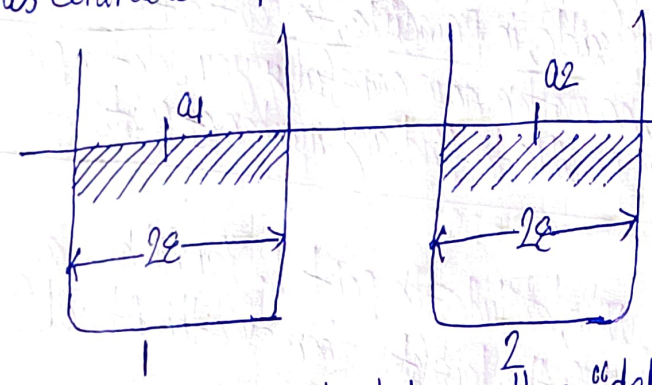
It means that we must have $|y_n - a_2| < \varepsilon$. Hence, we obtain that for all N greater than the largest among N_1 and N_2 the impossible must hold, namely, the terms of the sequence must simultaneously belong to the intervals 1 and 2. This contradiction proves the theorem.



$$\frac{2a_1+a_2}{3} < y_n < \frac{4a_1-a_2}{3}$$

$$a_1 - \frac{|a_1-a_2|}{3} < y_n < a_1 + \frac{|a_1-a_2|}{3}$$

$$a_2 - \frac{|a_1-a_2|}{3} < y_n < a_2 + \frac{|a_1-a_2|}{3}$$

$$\frac{4a_2-a_1}{3} < y_n < \frac{a_1+2a_2}{3}$$

This proof contains atleast two rather "delicate" points. Can you identify them?

Sam: Point 1: We must explicitly prove that the intervals 1 and 2 must be non-overlapping.

$$a_1 - \frac{|a_1-a_2|}{3} < y_n < a_1 + \frac{|a_1-a_2|}{3} \Rightarrow \frac{2a_1+a_2}{3} < y_n < \frac{4a_1-a_2}{3}$$

$$a_2 - \frac{|a_1-a_2|}{3} < y_n < a_2 + \frac{|a_1-a_2|}{3} \Rightarrow \frac{4a_2-a_1}{3} < y_n < \frac{a_1+2a_2}{3}$$

If $a_1 > a_2$

$$\boxed{\frac{4a_2-a_1}{3} < \frac{a_1+2a_2}{3} < \frac{2a_1+a_2}{3} < \frac{4a_1-a_2}{3}} \quad (1)$$

$$a_1 - \frac{|a_1-a_2|}{3} < y_n < a_1 + \frac{|a_1-a_2|}{3} \Rightarrow a_1 - \frac{a_2-a_1}{3} < y_n < a_1 + \frac{a_2-a_1}{3}$$

$$\Rightarrow \boxed{\frac{4a_1-a_2}{3} < y_n < \frac{2a_1+a_2}{3}}$$

$$a_2 - \frac{|a_1-a_2|}{3} < y_n < a_2 + \frac{|a_1-a_2|}{3} \Rightarrow a_2 - \frac{a_2-a_1}{3} < y_n < a_2 + \frac{a_2-a_1}{3}$$

$$\Rightarrow \boxed{\frac{2a_2+a_1}{3} < y_n < \frac{4a_2-a_1}{3}}$$

$$\boxed{\frac{4a_1-a_2}{3} < \frac{2a_1+a_2}{3} < \frac{2a_2+a_1}{3} < \frac{4a_2-a_1}{3}} \quad (2)$$

If $a_2 > a_1$

The relations (1) and (2) prove that Intervals 1 and 2 are non-overlapping.

Reader: I certainly notice one of them. If a_1 and a_2 are limits, no matter how the sequence behaves at the beginning, its terms in the long run have to concentrate simultaneously around a_1 and a_2, which is, of course, impossible. Why?

Author: Correct. But there is one more "delicate" point, namely, no matter how close a_1 and a_2 are, they should inevitably be spaced by a segment (a gap) of a small but definitely nonzero length.